UNIVERSITY OF RWANDA

Gako Campus

Computer Engineering | Year 4

MOBILE COMMUNICATIONS

Unit 2 Lab Report: Mini Projects (Lab 2)

Group Assignment

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Introduction

This report presents the work completed for the Unit 2 lab of the Mobile Communications course. It covers three mini projects that turn the theory of Unit 2 (radio propagation, digital modulation, OFDM and wireless network classes) into working Python simulations and data-driven comparisons.

Table 1. Overview of the mini projects

Project	Topic	Main outputs
2.1	Friis free-space path loss; BER of BPSK, QPSK, 16-QAM and 64-QAM over AWGN; link to Shannon-Hartley	Path-loss plots, BER curves, SNR gap at BER = 10^{-3}
2.2	OFDM transmitter/receiver with an adjustable cyclic prefix over a synthetic multipath channel	Constellations with/without CP, error rate vs CP length
2.3	Trade-off analysis of Bluetooth, Wi-Fi and WiMAX on range, throughput and power	Dataset, trade-off charts, scenario recommendations

Code repository

All source code, console outputs, generated figures and data files are stored in a GitHub repository. In line with the lab instructions, the code is not reproduced in this report: the report presents the method, the results and the interpretation, and the lecturer can inspect the implementation directly in the repository.

Table 2. Repository links

Module	Link
Lab 2 (all modules)	https://github.com/hirwagilbert/-Lab-2-Mobile-Communications
Mini Project 2.1	https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.1
Mini Project 2.2	https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.2
Mini Project 2.3	https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.3

Each module folder contains one Python script, the output of its last run, and its figures. The scripts use a fixed random seed, so every number reported below can be reproduced by running them. Python 3 with NumPy, SciPy, Matplotlib and pandas is sufficient; no SDR hardware is needed.

Mini Project 2.1: Free-Space Path Loss and Digital Modulation Performance

Objective

To build intuition for two central results of Unit 2, the Friis free-space path-loss equation and the Shannon-Hartley capacity theorem, by implementing them in Python and by using a Monte Carlo simulation to measure the bit error rate (BER) of BPSK and 16-QAM over an additive white Gaussian noise (AWGN) channel. QPSK and 64-QAM, channel coding, the log-distance model with shadowing, and a throughput comparison were added as the optional lab tasks.

Methodology

Part (a): Friis path loss

- Received power was computed as P_RX = P_TX + G_TX + G_RX − 20·log10(d) − 20·log10(f) − 32.44 (d in km, f in MHz), applied element-wise over a distance vector.
- Link parameters: P_TX = 43 dBm, G_TX = 15 dBi, G_RX = 0 dBi. Distance swept from 0.05 km to 10 km (500 points) at 900, 1900 and 3500 MHz.
- The 6 dB-per-doubling property was verified numerically by subtracting received powers one distance-doubling (or one frequency-doubling) apart.

Part (b): BER over AWGN

- A random bitstream of 399,996 bits was generated (random seed 0). This length is divisible by 1, 2, 4 and 6 bits per symbol, so every scheme uses exactly the same bits.
- Gray-coded mappers were implemented for BPSK, QPSK, 16-QAM and 64-QAM, normalised to unit average symbol energy.
- Noise variance was derived from the target Eb/N0 and the bits per symbol: N0 = 1 / (k · Eb/N0), with Es = 1. Eb/N0 was swept from 0 to 16 dB in 1 dB steps.
- Detection used the minimum-distance rule; BER = bit errors / total bits. BPSK was checked against the theoretical curve 0.5·erfc(√(Eb/N0)) before the other results were trusted.

Optional lab tasks

- Coding: a rate-1/2, constraint-length-3 convolutional encoder (generators 7, 5 in octal) with a hard-decision Viterbi decoder, simulated on 150,000 information bits with BPSK. Eb/N0 accounts for the code rate (Eb = Es / R).
- Propagation: log-distance model PL(d) = PL(d0) + 10·n·log10(d/d0) with d0 = 100 m at 900 MHz and n = 2, 3, 4; then log-normal shadowing with σ = 6 dB (five random realisations, n = 3).
- Throughput proxy: bits per symbol × (1 − BER) for each scheme, compared with the Shannon-Hartley curve log2(1 + SNR).

Results

Friis path loss and the 6 dB rule

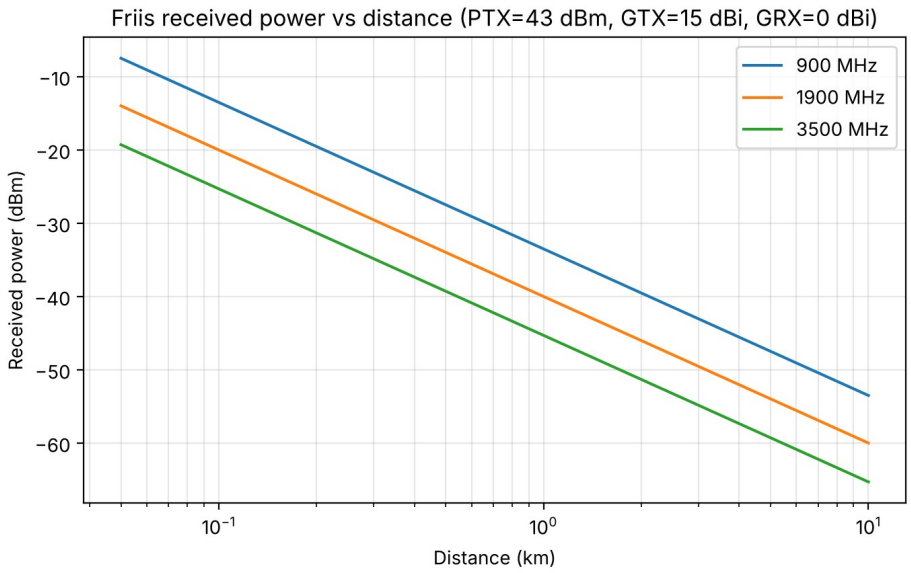


Figure 1. Friis received power versus distance at 900, 1900 and 3500 MHz (log distance axis).

Received power falls linearly against log(distance), and the higher the carrier frequency the lower the curve, at every distance. At 1 km, for example, the received power is −33.5 dBm at 900 MHz and −45.3 dBm at 3500 MHz. The 11.8 dB difference equals 20·log10(3500/900), as the equation predicts.

Table 3. Numerical check of the 6 dB-per-doubling relationship (printed by the script)

Test	Difference	Expected
P(1 km) − P(2 km) at 900 MHz	6.02 dB	6.02 dB
P(1 km) − P(2 km) at 1900 MHz	6.02 dB	6.02 dB
P(1 km) − P(2 km) at 3500 MHz	6.02 dB	6.02 dB
P(900 MHz) − P(1800 MHz) at 1 km	6.02 dB	6.02 dB
P(1900 MHz) − P(3800 MHz) at 1 km	6.02 dB	6.02 dB

BER versus Eb/N0

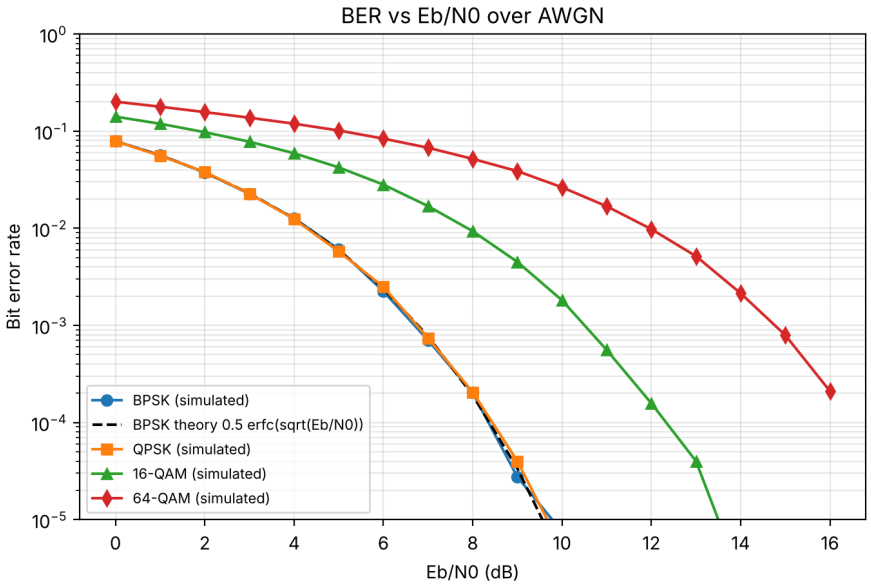


Figure 2. Simulated BER of BPSK, QPSK, 16-QAM and 64-QAM over AWGN, with the theoretical BPSK curve (dashed).

The simulated BPSK curve lies on top of the theoretical curve down to about 10^{-5}, which validates the noise scaling, the mapper and the detector. QPSK coincides with BPSK because it carries two independent BPSK streams on orthogonal axes, so each bit sees the same Eb/N0. Both QAM curves show the typical waterfall shape, shifted to the right.

Table 4. Selected BER values (399,996 bits; 0* means no errors observed, i.e. BER below about 2.5×10^{-6})

Eb/N0 (dB)	BPSK	QPSK	16-QAM	64-QAM	BPSK theory
0	7.87e-02	7.89e-02	1.41e-01	2.00e-01	7.86e-02
4	1.27e-02	1.24e-02	5.91e-02	1.19e-01	1.25e-02
6	2.24e-03	2.49e-03	2.81e-02	8.36e-02	2.39e-03
8	2.03e-04	2.05e-04	9.32e-03	5.17e-02	1.91e-04
10	7.50e-06	5.00e-06	1.81e-03	2.64e-02	3.87e-06
12	0*	0*	1.58e-04	9.78e-03	9.01e-09
16	0*	0*	0*	2.10e-04	2.27e-19

SNR gap at a target BER of 10^{-3}

The Eb/N0 needed to reach BER = 10^{-3} was read off each curve by interpolation in the log domain. Because the scheme carries k bits per symbol, the per-symbol SNR is Es/N0 = k · Eb/N0, and the table gives both. The last two columns compare each scheme with the Shannon limit for the same spectral efficiency η = k bits/s/Hz, which requires Eb/N0 ≥ (2^η − 1)/η.

Table 5. Required SNR at BER = 10^{-3} and distance from the Shannon limit

Scheme	Bits/ symbol	Eb/N0 (dB)	Es/N0 (dB)	Shannon min Eb/N0 (dB)	Gap to Shannon (dB)
BPSK	1	6.70	6.70	0.00	6.70
QPSK	2	6.75	9.76	1.76	4.99
16-QAM	4	10.50	16.52	5.74	4.76
64-QAM	6	14.77	22.55	10.21	4.56

Stated gap: at BER = 10^{-3}, 16-QAM needs 3.81 dB more Eb/N0 than BPSK, which is 9.83 dB more per-symbol SNR (Es/N0). 64-QAM needs 8.07 dB more Eb/N0 (15.85 dB more Es/N0) than BPSK.

Note on the lab manual's "8–10 dB" figure. The manual expects 16-QAM to need roughly 8–10 dB more than BPSK. The simulation reproduces this when the comparison is made in per-symbol SNR (9.83 dB). In Eb/N0, which normalises by bits per symbol, the gap is smaller (3.81 dB) because 16-QAM delivers four times as many bits for each unit of symbol energy. The report states both so that the numbers can be compared consistently.

Channel coding

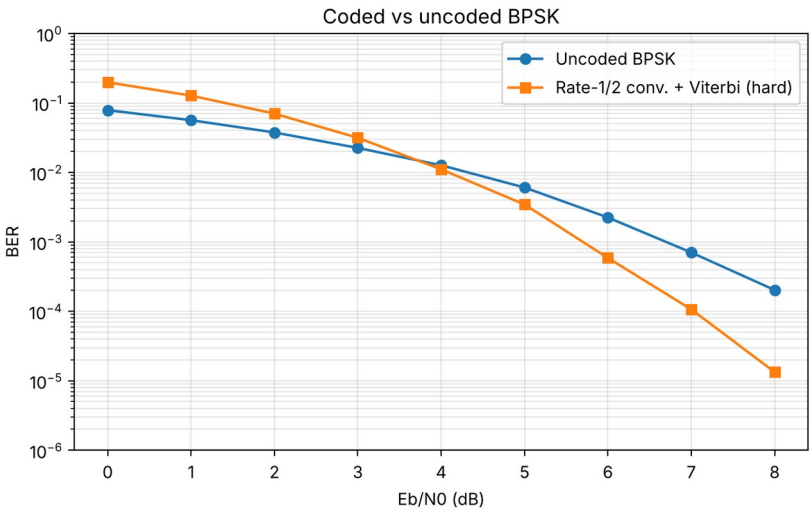


Figure 3. Uncoded BPSK versus rate-1/2 convolutional code with hard-decision Viterbi decoding.

At BER = 10^{-3} the coded system needs 5.70 dB Eb/N0 against 6.70 dB for uncoded BPSK, a coding gain of about 1.0 dB. The curves cross near 4 dB: below that the coded system is actually worse, because the decoder cannot correct the many channel errors. The gain is modest because the code is short (constraint length 3) and decoding is hard-decision.

Log-distance model and shadowing

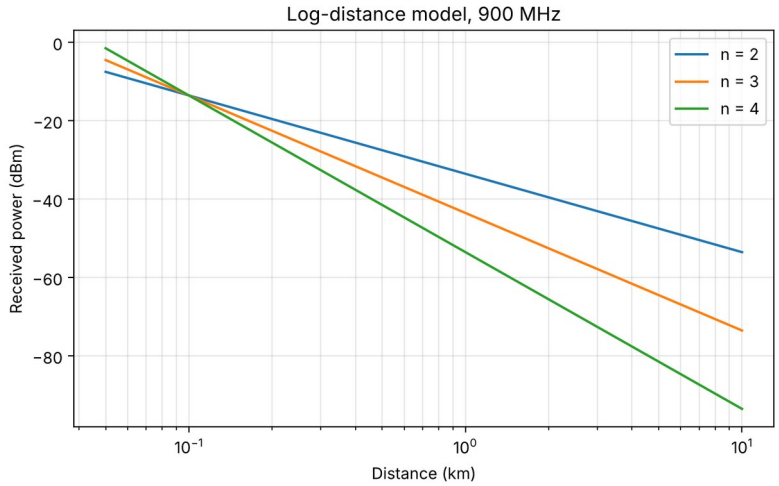


Figure 4. Log-distance model at 900 MHz for path-loss exponents n = 2, 3 and 4.

With d0 = 100 m the reference loss from Friis is 71.5 dB, and n = 2 reproduces the Friis result exactly (−33.5 dBm at 1 km). Each unit increase in n removes a further 10 dB per decade of distance beyond d0: at 1 km the received power is −33.5, −43.5 and −53.5 dBm for n = 2, 3 and 4.

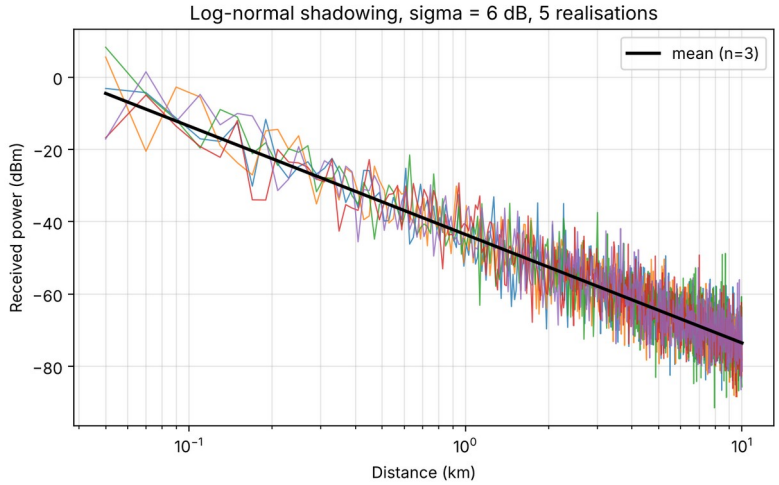


Figure 5. Five realisations of log-normal shadowing (σ = 6 dB) around the n = 3 mean path loss.

Shadowing adds a random, distance-independent offset. With σ = 6 dB most samples fall within ±12 dB (two standard deviations) of the mean, so two receivers at the same distance can differ by well over 10 dB. A link budget therefore needs a fade margin on top of the mean path loss.

Throughput proxy versus Shannon-Hartley

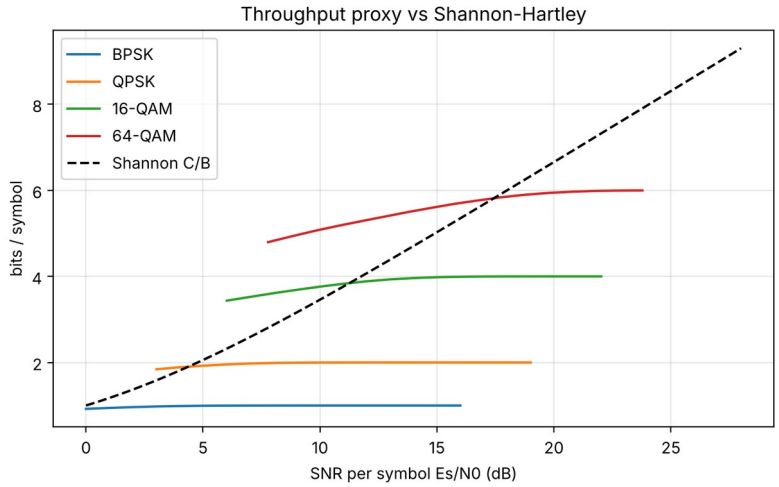


Figure 6. Throughput proxy k·(1 − BER) of each scheme compared with the Shannon-Hartley curve.

Each scheme's proxy saturates at its own bits-per-symbol ceiling (1, 2, 4 and 6), while the Shannon curve keeps rising with SNR. The proxy is only a rough indicator: it counts erroneous bits as delivered, which is why the 16-QAM and 64-QAM lines appear above the capacity curve at low SNR. A real error-free rate can never exceed capacity.

Discussion

The Shannon-Hartley theorem, C = B·log2(1 + SNR), gives the ceiling on error-free throughput for a given bandwidth and SNR. It does not predict the BER of a specific modulation, but it explains the direction and rough size of the gap that was measured. Raising spectral efficiency from 1 to 4 bits/s/Hz raises the minimum required Eb/N0 from 0 dB to 5.74 dB; the measured gap between BPSK and 16-QAM is 3.81 dB, and in per-symbol SNR it is 9.83 dB. A denser constellation also places its points closer together in signal space, so the same noise causes more decision errors.

The measured gap is consistent with Shannon-Hartley, but not numerically predicted by it, because the theorem describes an idealised capacity-achieving code with arbitrarily long blocks. Uncoded modulation always operates some distance below that ceiling. Table 5 shows this distance: every uncoded scheme needs between 4.6 and 6.7 dB more Eb/N0 than the Shannon limit at BER = 10^{-3}. Channel coding is the tool that narrows this distance, as the 1.0 dB gain of even a short convolutional code demonstrates.

Conclusion

The Friis equation was implemented and verified to give exactly 6.02 dB of extra loss per doubling of distance or frequency. The Monte Carlo BER simulation matched theory for BPSK and showed that QPSK costs no extra Eb/N0, while 16-QAM and 64-QAM need 3.81 dB and 8.07 dB more Eb/N0 (9.83 dB and 15.85 dB more Es/N0) than BPSK at BER = 10^{-3}. These gaps follow the trend set by the logarithmic SNR term in Shannon-Hartley, and the remaining distance to the capacity limit can be reduced by coding.

Code and outputs for this module: https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.1

Mini Project 2.2: OFDM Waveform Generator with a Cyclic Prefix Visualizer

Objective

To build a compact but functionally complete OFDM transmitter and receiver that generates a multi-subcarrier waveform with an IFFT, adds a cyclic prefix (CP), passes consecutive symbols through a synthetic multipath channel, and shows directly how the cyclic prefix removes inter-symbol interference (ISI).

System Design

Table 6. Simulation parameters

Parameter	Value
Subcarriers (N)	64
Modulation	Gray-coded QPSK, unit average symbol energy
OFDM symbols in the stream	400 (the first is discarded as a start-up transient)
CP lengths tested	0 to 16 samples
Noise	AWGN at 35 dB SNR, so that the good case is realistic rather than ideal
Receiver	Remove CP, FFT, one-tap zero-forcing equaliser using the known channel response
Metrics	Symbol error rate (SER) and mean-squared error (MSE) of recovered constellation points

The three-tap channel was convolved with the whole stream of back-to-back CP-OFDM symbols, so the tail of each symbol leaks into the start of the next. The same channel realisation and the same data were used for every CP length, so differences in the results come only from the CP.

Table 7. Synthetic multipath channel (gains normalised to unit energy)

Tap delay (samples)	Gain magnitude	Gain phase
0	0.811	0°
3	0.487	+51.6° (0.9 rad)
6	0.324	−120.3° (−2.1 rad)

The maximum excess delay is 6 samples, so a CP of at least 6 samples is needed to contain the ISI.

Verification steps

As the suggested approach requires, the building blocks were verified before the full chain was used. All three checks passed and are printed by the script:

- The QPSK mapper round-trips a known bit pattern, and its average symbol energy is 1.
- Applying the FFT to the noiseless IFFT output recovers the subcarrier values exactly (maximum error 4.0×10^{-16}, i.e. rounding error only).
- The first 16 samples of a CP-extended symbol are identical to its last 16 samples.

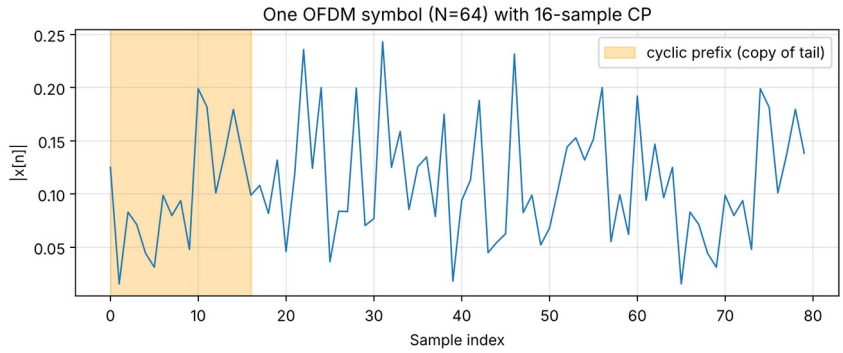


Figure 7. One OFDM symbol (N = 64) with a 16-sample cyclic prefix; the shaded region is a copy of the symbol's tail.

Results

Constellations with and without a sufficient CP

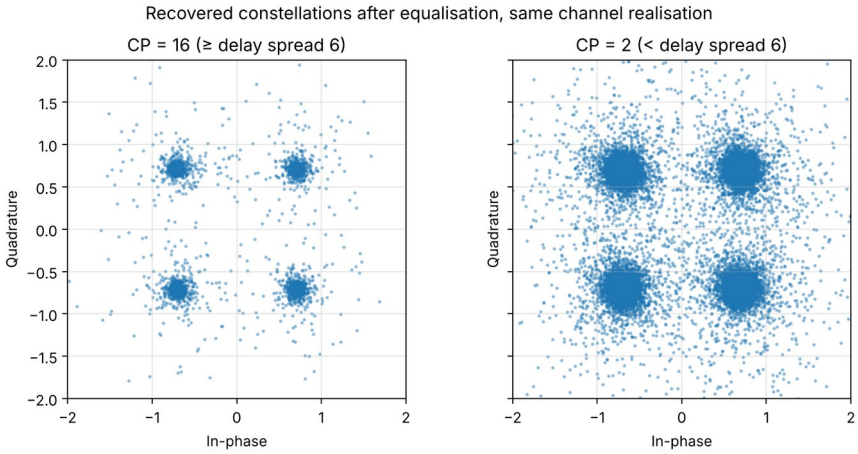


Figure 8. Recovered QPSK constellations on the same channel: CP = 16 (left) versus CP = 2 (right).

With CP = 16 the four constellation clusters are tight; the scattered points far from the clusters come from subcarriers in deep fades, where the equaliser amplifies noise. With CP = 2, which is shorter than the 6-sample delay spread, the clusters are heavily smeared and overlap the decision boundaries, because delayed energy from the previous symbol falls inside the FFT window and destroys subcarrier orthogonality.

Table 8. Error metrics for sufficient, short and absent CP

Case	CP length	SER	MSE
Sufficient CP	16	0.0017	7.78×10^{-3}
CP shorter than delay spread	2	0.0316	0.716
No CP	0	0.0488	1.59

Removing the CP raises the MSE by a factor of about 200 and the symbol error rate by a factor of about 29, compared with the sufficient-CP case.

Error rate versus CP length

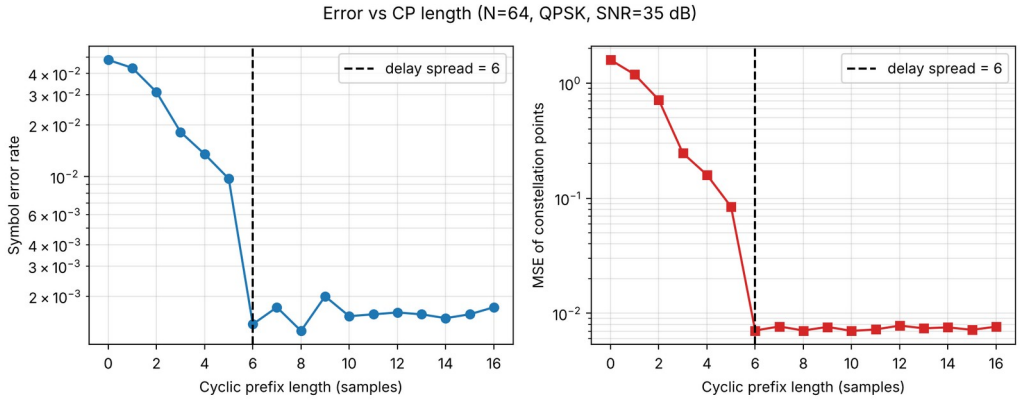


Figure 9. Symbol error rate (left) and MSE (right) versus CP length; the dashed line marks the 6-sample delay spread.

Table 9. Error versus CP length (selected values)

CP length (samples)	SER	MSE	CP overhead
0	0.0480	1.59	0%
2	0.0312	0.716	3.0%
4	0.0135	0.159	5.9%
5	0.0097	0.0842	7.2%
6	0.0014	7.07×10^{-3}	8.6%
8	0.0013	7.07×10^{-3}	11.1%
12	0.0016	7.81×10^{-3}	15.8%
16	0.0017	7.65×10^{-3}	20.0%

CP overhead is CP / (N + CP). Values in the SER column differ slightly from Table 8 because each CP length was run with fresh noise.

Discussion

The error rate falls steadily as the CP grows, drops sharply once the CP reaches 6 samples (the channel's delay spread), and then stays flat at about 1.5×10^{-3}. The behaviour follows from the two roles of the cyclic prefix.

- **Guard interval.** The first part of each received symbol is contaminated by the delayed tail of the previous symbol. If the CP is at least as long as the delay spread, all contaminated samples lie inside the CP, which the receiver discards. No energy from the previous symbol enters the FFT window, so there is no ISI.
- **Cyclic extension.** Because the CP is a copy of the symbol's tail, the channel's linear convolution with the transmitted signal looks like a circular convolution over the FFT window. A circular convolution becomes a simple multiplication by H[k] on each subcarrier after the FFT, so orthogonality is preserved and a one-tap equaliser per subcarrier is enough.

When the CP is shorter than the delay spread, neither condition holds: the window still contains part of the previous symbol, and the convolution is no longer circular, so subcarriers interfere with each other. The damage shrinks gradually as the CP grows because a longer CP absorbs more of the delayed paths, and it vanishes at CP = 6 where all paths are covered. Beyond that point the residual error is set by noise (zero-forcing equalisation amplifies noise on faded subcarriers), not by ISI, and extra CP only

wastes overhead: going from CP = 6 to CP = 16 raises overhead from 8.6% to 20% with no measurable improvement. A well-designed system therefore chooses a CP just longer than the expected delay spread.

Conclusion

A full OFDM chain (mapper, IFFT, CP insertion, multipath channel, CP removal, FFT and one-tap equalisation) was built and verified against a noiseless self-check. On a channel with a 6-sample delay spread, a sufficient CP gave a symbol error rate of 0.0017, while no CP gave 0.0488 and the MSE rose by a factor of about 200. The error-versus-CP sweep showed the expected threshold-and-plateau behaviour, with the threshold exactly at the delay spread.

Code and outputs for this module: https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.2

Mini Project 2.3: Comparing WPAN, WLAN, and WMAN Technologies

Objective

To build a small data-driven tool that captures the range, throughput and power trade-off across representative WPAN (Bluetooth), WLAN (Wi-Fi) and WMAN (WiMAX) technologies, to visualise that trade-off space, and to use it to recommend a technology for three deployment scenarios.

Dataset and Method

The dataset contains six entries covering the three network classes. Each entry records a typical range (m), a typical and a peak data rate (Mbps) and the transmit power (dBm). Two derived quantities were computed: a figure of merit, throughput × range (Mbps·m), and the transmit-power cost of throughput, in mW per Mbps. The script then applies each scenario's requirements to the table to test which technologies are feasible.

Table 10. Dataset used in the tool

Technology	Class	Band	Range (m)	Typical rate (Mbps)	Peak rate (Mbps)	Tx (dBm)
BLE 5 (LE 1M)	WPAN	2.4 GHz	50	1	2	0
Bluetooth Classic, Class 2	WPAN	2.4 GHz	10	2.1	3	4
Bluetooth Classic, Class 1	WPAN	2.4 GHz	100	2.1	3	20
Wi-Fi 5 (802.11ac)	WLAN	5 GHz	35	433	6933	20
Wi-Fi 6 (802.11ax)	WLAN	2.4/5 GHz	35	600	9608	20
Mobile WiMAX (802.16e)	WMAN	2.5/3.5 GHz	3000	4	30	23

Which values come from cited sources and which are engineering assumptions is set out below.

Sources and assumptions

- **Sourced:** Bluetooth power classes (Class 1: 100 mW / +20 dBm / ~100 m; Class 2: 2.5 mW / +4 dBm / ~10 m) [1]; BLE PHY rates of 1 and 2 Mb/s [2]; Wi-Fi 5 peak 6.933 Gbps and 433 Mbps for one stream at 80 MHz, Wi-Fi 6 peak 9.6 Gbps and 600 Mbps for one stream at 80 MHz, with no significant range increase of 802.11ax over 802.11ac [3]; mobile WiMAX 30 Mbps per subscriber in 10 MHz with typical rates of 3–5 Mbps, and an indoor Wi-Fi hotspot range of about 20 m [4].
- **Assumed:** BLE 1M typical range of 50 m (open, unobstructed); the typical rate of 2.1 Mbps and the 3 Mbps peak for Bluetooth Classic EDR; a Wi-Fi indoor range of 35 m (the cited 20 m is for a typical hotspot; 35 m is a middle value for modern indoor APs); a Wi-Fi transmit power of 20 dBm (regulatory limits are higher, but 15–20 dBm is common in practice); a mobile WiMAX cell range of 3 km and a transmit power of 23 dBm (a typical subscriber-station figure). The cited WiMAX source reports up to 50 km for line-of-sight fixed links, so 3 km is a conservative urban mobile figure.
- **Caveat:** range figures vary strongly with environment, antennas, obstructions and data rate, so the values are order-of-magnitude representative figures, not guarantees. Typical rates are used for scenario testing and peak rates are shown for reference only.

Results

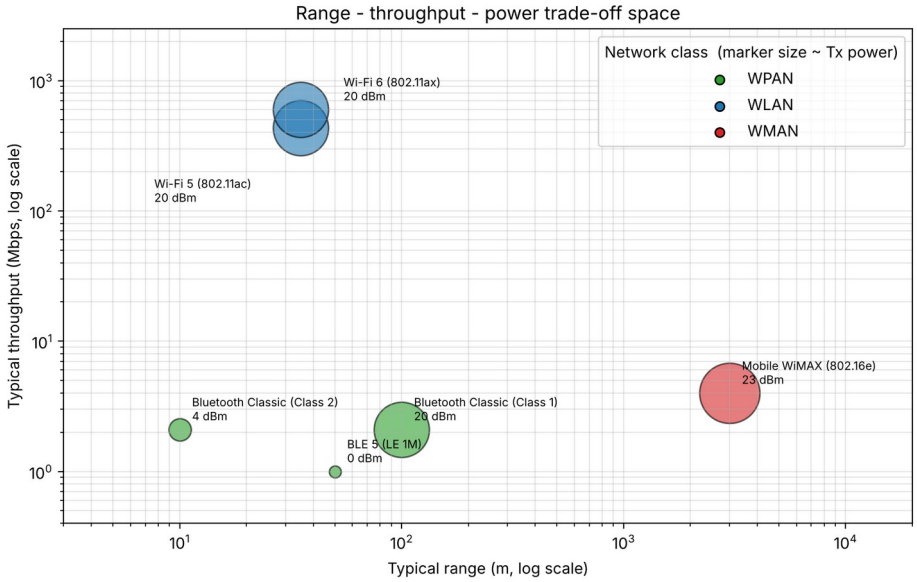


Figure 10. Range versus throughput (both log scale); marker size shows transmit power, colour shows network class.

The three network classes occupy distinct regions of the space. WPAN technologies sit at short range and low throughput with low power. WLAN is at short range but very high throughput (two orders of magnitude above WPAN). WMAN is the only class reaching kilometre range, at moderate throughput. No technology is strong on all three axes; moving up in range or throughput generally costs transmit power.

Table 11. Derived figures of merit

Technology	Rate × range (Mbps·m)	Tx power (mW)	mW per Mbps
BLE 5 (LE 1M)	50	1.0	1.00
Bluetooth Classic, Class 2	21	2.5	1.20
Bluetooth Classic, Class 1	210	100	47.6
Wi-Fi 5 (802.11ac)	15,155	100	0.23
Wi-Fi 6 (802.11ax)	21,000	100	0.17
Mobile WiMAX (802.16e)	12,000	200	49.9

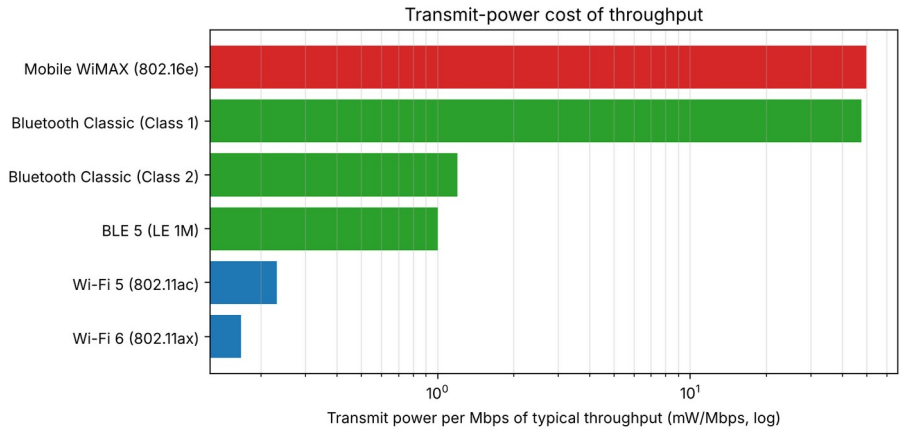


Figure 11. Transmit power spent per Mbps of typical throughput (log scale).

Wi-Fi is by far the cheapest in transmit power per Mbps (0.17–0.23 mW/Mbps) because its throughput is so high, but its absolute transmit power of 100 mW is not suitable for coin-cell devices. BLE has the lowest absolute power (1 mW) and low power per Mbps among the low-rate technologies. WiMAX and Class 1 Bluetooth are the most expensive per Mbps, about 48–50 mW/Mbps, since they spend power on range rather than rate.

Deployment Scenarios

Each scenario has numbered requirements. The script tests every technology against them, and a technology is feasible only if it passes all of range, typical throughput and transmit-power limit.

Table 12. Scenario requirements and feasibility result

Scenario	Requirements	Feasible	Recommendation
S1 Smart home sensors	(1) 30 coin-cell sensors; (2) coverage ≥ 20 m; (3) ≥ 0.1 Mbps per node; (4) Tx ≤ 4 dBm	BLE only	BLE 5
S2 Campus network	(1) ≥ 100 Mbps per user; (2) cell radius ≥ 30 m; (3) mains-powered APs; hundreds of users	Wi-Fi 5, Wi-Fi 6	Wi-Fi 6
S3 Metro fixed link	(1) distance ≈ 3 km; (2) ≥ 4 Mbps per site; (3) mains power at both ends	WiMAX only	Mobile WiMAX

Table 13. Which requirement each technology fails (– means all met)

Technology	S1	S2	S3
BLE 5 (LE 1M)	–	rate	range, rate
Bluetooth Classic, Class 2	range	range, rate	range, rate
Bluetooth Classic, Class 1	power	rate	range, rate
Wi-Fi 5 (802.11ac)	power	–	range
Wi-Fi 6 (802.11ax)	power	–	range
Mobile WiMAX (802.16e)	power	rate	–

S1: Smart home with many low-power sensors

BLE is the only technology that passes all four requirements: its 50 m range exceeds the 20 m needed, its 1 Mbps is ten times the 0.1 Mbps per node, and its 0 dBm (1 mW) transmit power is inside the 4 dBm budget. Bluetooth Classic Class 2 meets the power limit but its 10 m range is too short for a whole home. Class 1 Bluetooth and Wi-Fi transmit at 20 dBm (100 mW), which is 16 dB (40 times) above the budget, and WiMAX at 23 dBm is higher still. Secondary factors reinforce the choice: BLE chips are inexpensive, use licence-free 2.4 GHz spectrum, and support mesh networking. The main complication is that Wi-Fi shares the 2.4 GHz band, so interference must be managed, and real battery life depends on the duty cycle rather than only on transmit power.

S2: Campus network with wide coverage and high aggregate throughput

Both Wi-Fi standards meet the range (35 m against 30 m) and the 100 Mbps per-user requirement (433 and 600 Mbps typical per link). The other technologies fail on rate: BLE and Bluetooth Classic offer at most 2.1 Mbps, and WiMAX 4 Mbps typical, 30 Mbps peak shared by all users of a cell. Wi-Fi 6 is recommended because its typical rate is 39% higher than Wi-Fi 5 (600 against 433 Mbps), it works in both the 2.4 and 5 GHz bands, and it adds features aimed at dense deployments such as OFDMA. Since neither standard extends the 35 m range significantly, campus-wide coverage must come from many overlapping access points on a wired backbone, which increases cost and requires channel planning. Mains power removes the power constraint.

S3: Metro-area fixed wireless link spanning several kilometres

Only WiMAX reaches the required 3 km (3000 m). Wi-Fi's 35 m would need roughly 85 access-point hops to span the same distance, which is impractical, and Bluetooth reaches at most 100 m. WiMAX's typical 4 Mbps meets the 4 Mbps per-site requirement exactly, with 30 Mbps peak giving some headroom, at 23 dBm transmit power that mains-powered sites can afford. Secondary factors complicate the answer: WiMAX commonly uses licensed spectrum (2.5/3.5 GHz), which adds cost and regulatory steps, capacity is shared among users of a cell, and in practice LTE and 5G fixed wireless have largely replaced WiMAX in new deployments. WiMAX is used here as the representative WMAN technology of Unit 2, and the same range-first reasoning applies to its successors.

Conclusion

The comparison shows that range, throughput and power cannot all be maximised together: WPAN trades throughput and range for very low power, WLAN trades range and power for throughput, and WMAN trades throughput for kilometre-scale range. Applying numeric requirements to the dataset gave a clear, unique recommendation for each scenario: BLE for the smart home, Wi-Fi 6 for the campus, and WiMAX for the metro link. The main limitation is that range and power values depend on the environment and are partly assumptions, which is stated in the dataset notes.

Code, dataset and outputs for this module: https://github.com/hirwagilbert/-Lab-2-Mobile-Communications/tree/main/Mini-Project-2.3

Overall Conclusion

The three mini projects connect the theory of Unit 2 to measurable results.

- **Propagation and capacity (2.1):** Friis path loss grows by exactly 6.02 dB per doubling of distance or frequency; BPSK and QPSK need 6.7 dB Eb/N0 for a BER of 10^{-3}, while 16-QAM and 64-QAM need 10.5 dB and 14.8 dB. These gaps follow the logarithmic SNR term of Shannon-Hartley, and uncoded schemes remain 4.6–6.7 dB from the capacity limit.
- **OFDM (2.2):** a cyclic prefix at least as long as the channel's delay spread removes ISI and preserves subcarrier orthogonality; with a 6-sample delay spread the error rate dropped sharply at CP = 6 and then stayed flat, so longer prefixes only add overhead.
- **Network technologies (2.3):** the choice between WPAN, WLAN and WMAN follows from the range, throughput and power requirements, and a simple dataset-driven filter produced a unique justified choice for each scenario.

All code, console output, figures and data are available in the repository listed below for verification.

Repository: https://github.com/hirwagilbert/-Lab-2-Mobile-Communications

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